MOHAMMAD SOHAIL AHMED

BACKEND SOFTWARE ENGINEER | JAVA | SPRING BOOT | AWS | SAP COMMERCE CLOUD

Hyderabad, India | +91 81066 37318 | sohailahmed.mohammad01@gmail.com
linkedin.com/in/mohammad-sohail-ahmed | github.com/sohailahmedmohammad01 | sohailahmedmohammad01.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend Software Engineer with nearly 3 years of experience building enterprise commerce platforms for Changi Airport Group in Java, Spring Boot, AWS, and SAP Commerce Cloud. Currently leading TREX-side delivery of Singapore's Beverage Container Return Scheme (BCRS) across the TREX marketplace and iShopChangi, from business requirements through pricing, order, refund, and invoice flows. Sole owner of the DocuSign merchant onboarding flow that has brought 200+ merchants onto TREX. Also a freelance backend developer on a healthcare information system, and builder of applied-AI systems with RAG and multi-agent orchestration.

PROFESSIONAL EXPERIENCE

CJSS Technologies Pvt. Ltd.
Hyderabad, India

Software Engineer
Jan 2024 - Present

Client: Changi Airport Group (CAG) | TREX marketplace platform and iShopChangi (ISC) e-commerce on SAP Commerce Cloud

Beverage Container Return Scheme (BCRS) - TREX and iShopChangi

- Lead TREX-side design and development of Singapore's Beverage Container Return Scheme across TREX and iShopChangi, covering every eligible beverage, wine, and pharmacy product and all Changi merchants; go-live scheduled for 10 Nov 2026.
- Built container-deposit handling end to end: eligible-product flagging, deposit as a separate price component and order line item, refund logic, invoice changes, and TREX-to-SAP Commerce data synchronization.
- Gathered and refined requirements directly with business stakeholders and presented the end-to-end solution demo to the business team, earning recognition for the solution and the team's delivery.

TREX - Cloud-Native Marketplace Platform for iShopChangi

- Single-handedly designed, integrated, and delivered DocuSign digital-signature merchant onboarding end to end; 200+ merchants have onboarded to TREX through digitized agreements with pre-populated terms and values, earning recognition from Changi Airport Group.
- Integrated TREX with SAP Commerce Cloud for iShopChangi, covering catalog synchronization, pricing, promotions, and order data exchange.
- Build and maintain Java/Spring Boot microservices, REST and GraphQL APIs, and marketplace business logic for product, pricing, order, merchant, contract, user, and promotion workflows.
- Build AWS Lambda/AppSync serverless workflows and Dockerized ECS services on DynamoDB and relational databases; integrated Okta for authentication.
- Ran a hands-on Datadog vs Dynatrace proof of concept and recommended the platform for simpler, centralized application logging and monitoring.
- Resolve production defects across backend services, integrations, and database operations, tracing backend and database bottlenecks with Datadog and Dynatrace.
- Write JUnit and Mockito tests, support regression testing, and validate releases before production; apply AI-assisted test generation and code review.
- Mentor junior engineers through code reviews and debugging sessions.

iShopChangi (ISC) - E-commerce on SAP Commerce Cloud

- Build OCC APIs, custom extensions, items.xml models, and populators/converters from business requirements across catalog, pricing, promotions, checkout, and order flows.
- Work with FlexibleSearch, ImpEx, Solr, CronJobs, catalog synchronization, and Backoffice to deliver and support commerce processes.
- Troubleshoot cross-layer incidents between TREX and SAP Commerce and validate releases.

Smart Softech Solutions **Freelance**

Backend Developer **Jun 2026 - Present**

Project: SmartHIS - Healthcare Information System | Java | Spring Boot | PostgreSQL | JPA/Hibernate | JWT/RBAC

- Build the modular backend for patient registration, appointments, billing, pharmacy, laboratory, administration, and audit logging.
- Secure access with JWT authentication and role-based access control across modules.
- Design relational data models, validations, exception handling, and audit trails; handle deployment troubleshooting, defect resolution, and enhancements.

SELECTED ENGINEERING PROJECTS

Zaiqa - Hospitality Operations Platform **Building Now**

Java 21 | Spring Boot | Spring Modulith | PostgreSQL | Flyway | React | TypeScript

- Building a multi-site platform that runs hotel PMS and restaurant operations (rooms, stays, folios, menus, orders, tables, housekeeping, inventory, billing, and reporting) from one system.
- Designed a Spring Boot modular monolith of 14 domain modules with boundaries verified by Spring Modulith, branch-scoped JWT and RBAC across seven roles, and Flyway-managed PostgreSQL.
- Built a React and TypeScript operations UI on top of the backend APIs.

Multi-Agent RAG Platform **Independent Project**

Spring Boot | Python FastAPI | Gemini | Qdrant | Vector Search

- Implemented an end-to-end RAG pipeline covering document ingestion, embeddings, semantic retrieval, and grounded response generation across Spring Boot and FastAPI services.
- Applied Qdrant vector search, metadata/payload filtering, and role-based access control to deliver secure, context-aware answers across specialized agents.

TECHNICAL SKILLS

Languages & UI: Java, SQL, Python, JavaScript, TypeScript, React

Backend: Spring Boot, Spring MVC, Spring Security, Spring Modulith, JPA/Hibernate, REST APIs, GraphQL, Microservices, JWT/RBAC, Okta, DocuSign

SAP Commerce Cloud: TREX-SAP Integration, OCC APIs, Custom Extensions, items.xml, FlexibleSearch, ImpEx, Solr, CronJobs, Catalog Synchronization, Backoffice

Cloud & Data: AWS Lambda, AppSync, ECS, EC2, S3, CloudWatch, DynamoDB, PostgreSQL, MySQL, Flyway

Delivery & Observability: Git, Maven, Docker, Jenkins, CI/CD, Dynatrace, Datadog, JUnit, Mockito, Integration Testing, Root-Cause Analysis, Code Review

AI: RAG, Qdrant, Vector Search, Embeddings, Gemini, Multi-Agent Systems, FastAPI, AI-Assisted Testing and Code Review

EDUCATION

Rajiv Gandhi University of Knowledge Technologies (IIIT-RGUKT), Basar **2020 - May 2024**

B.Tech, Computer Science & Engineering | GPA: 8.9/10

CERTIFICATIONS & RECOGNITION

CJSS Best Employee Phoenix Award 2026 - Recognized for AI-driven innovation and client delivery.
Client recognition, Changi Airport Group - For single-handedly delivering DocuSign merchant onboarding on TREX.
Oracle Certified Associate, Java SE 8 Programmer - Issued Jul 2025 | Credential ID 102042221OCAJSE8
AWS Certified Cloud Practitioner - Amazon Web Services
Courses: Data Science (Pregrad, Oct 2022) | Python Core | Blockchain and Its Applications (NPTEL)